SUPPLEMENTARY MATERIAL

VIII. NOTATIONS AND MATHEMATICAL DEFINITIONS

In this section, we introduce some symbol definitions and the preliminary knowledge, and supplement the specific definitions of related concepts mentioned in Section II.

A. Definitions of norm

We first introduce the definitions of norm. Given a matrix $A \in \mathbb{R}^{m_1 \times m_2}$, let $a_{i_1 i_2}$ denote the (i_1, i_2)-th component of A, then the Frobenius norm of A is defined as

$$\|A\|_F := \sqrt{\sum_{i_1=1}^{m_1} \sum_{i_2=1}^{m_2} a_{i_1 i_2}^2}.$$

then the ℓ_1-norm of A is defined as

$$\|A\|_1 := \sum_{i_1=1}^{m_1} \sum_{i_2=1}^{m_2} |a_{i_1 i_2}|.$$

B. Mathematical definitions for nonconvex analysis

We first define $\langle \cdot, \cdot \rangle$ as the inner product.

Next, we recall several basic concepts for nonconvex analysis [1], [2].

Definition 1. *Let $\boldsymbol{S}$ be a set, the derived set $\boldsymbol{S}'$ is defined as the set of overall cluster points of $\boldsymbol{S}$.*

Proposition 1. *A bounded closed set in finite-dimensional space is a compact set.*

Definition 2. *A function $g : \mathbb{R}^n \to (-\infty, +\infty]$ is said to be proper if $\mathrm{dom}\ g \neq \emptyset$, where $\mathrm{dom}\ g = \{x \in \mathbb{R} :\ g(x) < \infty\}$.*

Definition 3. *If $\forall x_k \in \mathrm{dom}\ f$, we have*

$$\lim_{k\to\infty} x_k = x,\ f(x) \leq \liminf_{k\to\infty} f(x_k)\ ,$$

then f is said to be lower semicontinuous on $\mathrm{dom}\ f$.

Definition 4. *Let f be a proper lower semicontinuous function, the Fréchet subdifferential of f at x, written $\hat{\partial} f(x)$, is the set of all vectors u which satisfy*

$$\lim_{y\neq x, y\to x} \cdot \frac{f(y)-f(x)-\langle u,\ y-x\rangle}{\|y-x\|} \geq 0,$$

when $x \notin \mathrm{dom}\ f$, then set $\hat{\partial} f(x) = \emptyset$.

Proposition 2. *If f is a proper lower semicontinuous function and has a local minimum at x^*, then $0 \in \partial f(x^*)$.*

Proposition 3. *Let f be a proper lower semicontinuous function, and g be a continuously differential function, then we have $\forall x \in \mathrm{dom}\ f$, $\partial(f+g)(x) = \partial f(x) + \nabla g(x)$.*

Proposition 4. *Set f: $R^n \to R$, if $\|\nabla f(x) - \nabla f(y)\| \leq L\|x-y\|$, then $f(y) \leq f(x) + \langle \nabla f(x), y-x\rangle + \frac{L}{2}\|y-x\|^2$, where L is the Lipschitz constant of ∇f.*

Definition 5. *Desingularization function: we denote the desingularization function $\phi : [0, \eta) \to \mathbb{R}_+$ which satisfies these conditions: (i) $\phi(0) = 0$. (ii) ϕ is continuously differentiable on $[0, \eta)$, and ϕ is a concave function from in $(0, \eta)$. (iii) $\phi(s)' > 0$ $(\forall s \in (0, \eta))$.*

Definition 6. *Let f be a proper lower semicontinuous function, f is said to have the KŁ property on $\bar{u} \in dom(\partial f)$, if there exists $\eta \in (0, +\infty]$, and U is the neighborhood of $\bar{u}$, any u in U that satisfies the condition $f(\bar{u}) < f(u) < f(\bar{u}) + \eta$ has the following inequalities established.*

$$\phi'(f(u) - f(\bar{u}))dist(0, \partial f(u)) > 1,$$

where ϕ satisfy (Definition 5). If f has the KŁ property at each point of $\mathrm{dom}\ \partial f$, then f is a KŁ function.

Proposition 5. *(Uniformized KŁ property) Let Ω be a compact set and f: $\mathbb{R}^n \to \mathbb{R} \cup \{\infty\}$ be a proper and lower semicontinuous function. Assume that f is constant on Ω and is a KŁ function. Then, there exist $\epsilon > 0$, $\eta > 0$ such that $\forall x \in \Omega$ and all x in the intersection $\{x \in \mathbb{R}^n : dist(x, \Omega) < \epsilon\} \cap \{f(\bar{x}) < f(x) < f(\bar{x}) + \eta\}$ one has*

$$\phi'(f(x) - f(\bar{x}))dist(0, \partial f(x)) > 1.$$

C. Graph Laplacian regularization

Given a matrix $X \in \mathbb{R}^{m_1 \times R}$, where R is the sample size, let $W = \{w_{ij},\ i, j = 1, 2.....R\}$ be the weight matrix with heat kernel weight, then w_{ij} can be calculated as follows.

$$w_{ij} = \begin{cases} e^{\frac{\|X_{i,:} - X_{j,:}\|_F^2}{-\sigma^2}} & \text{if } X_{j,:} \in \mathcal{N}_p(X_{i,:})\ or X_{i,:} \in \mathcal{N}_p(X_{j,:}), \\ 0 & \text{else,} \end{cases} \quad (9)$$

where $X_{i,:}$ is the i-th row of X, $\mathcal{N}_p(X_{i,:})$ is the set of p samples closest to $X_{i,:}$ in the graph, and σ^2 denotes the divergence. We set $\sigma^2 = 1$ for our method.

IX. CONVERGENCE ANALYSIS

In this section, we prove all the theorems in Section V. Notably, to simplify the proof in this section, we have already defined the sequence generated by Algorithm 1 in the main text as

$$z^k = A_{(N)}^k,\ c^k = Y_{(N)}^k. \quad (10)$$

A. Proof of Theorem 1

Proof. Because

$$\begin{aligned} &\nabla_{A_i} H(\{A_j\}_{j=1}^{i-1}, B_i, \{A_j\}_{j=i+1}^N) \\ &- \nabla_{A_i} H(\{A_j\}_{j=1}^{i-1}, C_i, \{A_j\}_{j=i+1}^N) \\ &= ((\prod_{j=1}^{i-1} A_j)^T \prod_{j=1}^{i-1} A_j)(B_i - C_i)((\prod_{j=i+1}^{N} A_j)(\prod_{j=i+1}^{N} A_j)^T). \end{aligned}$$

From the Frobenius norm is a self-consistent matrix norm, which conforms to the norm definition of a linear operator [1], thus we have

$$\begin{aligned} &\|\nabla_{A_i} H(\{A_j\}_{j=1}^{i-1}, B_i, \{A_j\}_{j=i+1}^N) \\ &- \nabla_{A_i} H(\{A_j\}_{j=1}^{i-1}, C_i, \{A_j\}_{j=i+1}^N)\|_F \\ &= \|((\prod_{j=1}^{i-1} A_j)^T \prod_{j=1}^{i-1} A_j)(B_i - C_i)(\prod_{j=i+1}^{N} A_j)(\prod_{j=i+1}^{N} A_j)^T\|_F \\ &\leq \|(\prod_{j=1}^{i-1} A_j)^T \prod_{j=1}^{i-1} A_j\|_F \\ &\times \|(B_i - C_i) \prod_{j=i+1}^{N} A_j(\prod_{j=i+1}^{N} A_j)^T)\|_F \\ &\leq \|(\prod_{j=1}^{i-1} A_j)^T \prod_{j=1}^{i-1} A_j\|_F \times \| \prod_{j=i+1}^{N} A_j(\prod_{j=i+1}^{N} A_j)^T)\|_F \\ &\times \|(B_i - C_i)\|_F. \end{aligned}$$

The Theorem 1 is true. □

B. Proof of Theorem 2

Proof. (i) If $J(A_{(N)}^{k+1}) \leq J(A_{(N)}^{k}) - \rho_t \|A_{(N)}^{k+1} - Y_{(N)}^{k}\|_F^2$, then the Theorem 2(i) is true. If not, since Theorem 1, we know that $H(A_{(N)})$ satisfies Proposition 4. Thus, we have

$$H(A_1^{k+1}, \{A_i^k\}_{i=2}^N) \leq \langle \nabla_{A_1} H(A_{(N)}^k), A_1^{k+1} - A_1^k \rangle + \frac{L_{\nabla_{A_1^k} H}}{2} \|A_1^{k+1} - A_1^k\|_F^2 + H(A_{(N)}^k). \tag{11}$$

From Eq. (8), we can also obtain

$$F_1(A_1^k) \geq F_1(A_1^{k+1}) + \frac{1}{2\sigma_1^k} \|A_1^{k+1} - A_1^k\|_F^2 + \langle \nabla_{A_1} H(A_{(N)}^k), A_1^{k+1} - A_1^k \rangle. \tag{12}$$

Then sum of the Eq. (11) and Eq. (12), we have

$$H(A_1^{k+1}, \{A_i^k\}_{i=2}^N) + F_1(A_1^{k+1}) \leq H(A_{(N)}^k) + F_1(A_1^k) - \rho \|A_1^{k+1} - A_1^k\|_F^2, \tag{13}$$

where $\rho = \min(\frac{1}{2\sigma_1^k} - \frac{L_{\nabla_{A_1^k} H}}{2})$. Assuming Theorem 2(i) holds when $n = N - 1$, i.e.,

$$H(A_{(N-1)}^{k+1}, A_N^k) + \sum_{i=1}^{N-1} F_i(A_i^{k+1}) \leq H(A_{(N)}^k) + \sum_{i=1}^{N-1} F_i(A_i^k) - \rho \|A_{(N-1)}^{k+1} - A_{(N-1)}^k\|_F^2, \tag{14}$$

where $\rho = \min(\{\frac{1}{2\sigma_i^k} - \frac{L_{\nabla_{A_i^k} H}}{2}\}_{i=1}^{N-1})$. Similarly, from Eq. (8), we infer

$$F_N(A_N^k) \geq F_N(A_N^{k+1}) + \frac{1}{2\sigma_N^k} \|A_N^{k+1} - A_N^k\|_F^2 + \langle \nabla_{A_N} H(A_{(N-1)}^{k+1}, A_N^k), A_N^{k+1} - A_N^k \rangle. \tag{15}$$

From Theorem 1 and Proposition 4, we also have

$$H(A_{(N)}^{k+1}) \leq H(A_{(N-1)}^{k+1}, A_N^k) + \langle \nabla_{A_N} H(A_{(N-1)}^{k+1}, A_N^k), A_N^{k+1} - A_N^k \rangle + \frac{L_{\nabla_{A_N^k} H}}{2} \|A_N^{k+1} - A_N^k\|_F^2. \tag{16}$$

Then sum of the Eq. (14), Eq. (15) and Eq. (16), we have

$$J(A_{(N)}^{k+1}) \leq J(A_{(N)}^k) - \rho \|z^{k+1} - c^k\|_F^2, \tag{17}$$

where $\rho = \min(\{\frac{1}{2\sigma_i^k} - \frac{L_{\nabla_{A_i^k} H}}{2}\}_{i=1}^N)$. This shows the Theorem 2(i) is true.

(ii) We know that $\frac{\alpha}{2} \operatorname{Tr}((A_N)^T L A_N)$ in Eq. (6) is nonnegative. Hence, Eq. (6) is nonnegative. From Eq. (17), we have

$$\rho \|z^{k+1} - c^k\|_F^2 \leq J(A_{(N)}^k) - J(A_{(N)}^{k+1}).$$

Sum of both side, since Eq. (6) is nonnegative, thus we have

$$\begin{aligned} \rho \sum_{k=1}^{\infty} \|z^{k+1} - c^k\|_F^2 &\leq \sum_{k=1}^{\infty} (J(A_{(N)}^k) - J(A_{(N)}^{k+1})) \\ &= J(A_{(N)}^1) - \inf J \\ &< \infty. \end{aligned}$$

It follows that

$$\lim_{k \to \infty} \|z^{k+1} - c^k\|_F^2 = 0.$$

□

C. Lemma: subgradient lower bound property of Algorithm 1

Before continuing with the proof of the theorems of Section V, we prove the following lemma.

Lemma 1. *If we define*

$$f_i^{k+1} = \nabla_{A_i} H(\{A_j^{k+1}\}_{j=1}^i, \{A_j^k\}_{j=i+1}^N) + \frac{1}{\sigma_i^k}(Y_i^k - A_i^{k+1}) - \nabla_{A_i} H(\{A_j^{k+1}\}_{j=1}^{i-1}, Y_i^k, \{A_j^k\}_{j=i+1}^N). \tag{18}$$

Then we have $\|\{f_i^{k+1}\}_{i=1}^N\|_F \leq \rho_b \|z^{k+1} - c^k\|_F$, *where* ρ_b *is a positive constant.*

Proof. By Proposition 2 and Proposition 3, Eq. (8) follows that

$$0 \in \partial_{A_i} F_i(A_i^{k+1}) - \frac{1}{\sigma_i^k}(Y_i^k - A_i^{k+1}) + \nabla_{A_i} H(\{A_j^{k+1}\}_{j=1}^{i-1}, Y_i^k, \{A_j^k\}_{j=i+1}^N). \tag{19}$$

Thus we also have

$$\begin{aligned} f_i^{k+1} &= \nabla_{A_i} H(\{A_j^{k+1}\}_{j=1}^i, \{A_j^k\}_{j=i+1}^N) - \frac{1}{\sigma_i^k}(A_i^{k+1} - Y_i^k) \\ &- \nabla_{A_i} H(\{A_j^{k+1}\}_{j=1}^{i-1}, Y_i^k, \{A_j^k\}_{j=i+1}^N) \\ &\in \nabla_{A_i} H(\{A_j^{k+1}\}_{j=1}^{i-1}, A_i^{k+1}, \{A_j^k\}_{j=i+1}^N) \\ &+ \partial_{A_i}(\sum_{j=1}^i F_j(A_j^{k+1}) + \sum_{j=i+1}^N F_j(A_j^k)) \\ &= \partial_{A_i} J(\{A_j^{k+1}\}_{j=1}^{i-1}, A_i^{k+1}, \{A_j^k\}_{j=i+1}^N). \end{aligned} \tag{20}$$

Since Theorem 1, we have

$$\begin{aligned} \|f_i^{k+1}\|_F &= \|\nabla_{A_i} H(\{A_j^{k+1}\}_{j=1}^i, \{A_j^k\}_{j=i+1}^N) + \frac{1}{\sigma_i^k}(Y_i^k - A_i^{k+1}) \\ &- \nabla_{A_i} H(\{A_j^{k+1}\}_{j=1}^{i-1}, Y_i^k, \{A_j^k\}_{j=i+1}^N)\|_F, \\ &\leq \|\nabla_{A_i} H(\{A_j^{k+1}\}_{j=1}^i, \{A_j^k\}_{j=i+1}^N) \\ &- \nabla_{A_i} H(\{A_j^{k+1}\}_{j=1}^{i-1}, Y_i^k, \{A_j^k\}_{j=i+1}^N)\|_F \\ &+ \|\frac{1}{\sigma_i^k}(A_i^{k+1} - Y_i^k)\|_F \\ &\leq (L_{\nabla_{A_i^k} H} + \frac{1}{\sigma_i^k}) \|A_i^{k+1} - Y_i^k\|_F. \end{aligned} \tag{21}$$

Thus we infer

$$\|\{f_i^{k+1}\}_{i=1}^N\|_F \leq \rho_b \|z^{k+1} - c^k\|_F, \tag{22}$$

where $\rho_b = \max(\{\frac{1}{\sigma_i^k} + L_{\nabla_{A_i^k} H}\}_{i=1}^N)$. □

D. Proof of Theorem 3

Proof. (i) Observe that z' can be viewed as an intersection of compact sets

$$z' = \cap_{q \in \mathbb{N}} \overline{\cup_{k \geq q} \{z^k\}}.$$

From proposition 1, z' is compact.

(ii) $\forall \overline{z} \in z'$, there exists a subsequence z^{k_j} such that

$$\lim_{j \to \infty} z^{k_j} = \overline{z}.$$

Let

$$F(z^{k_j}) = \sum_{i=1}^N F_i(A_i^{k_j}). \tag{23}$$

Since F_i is lower semicontinuous [2], from Definition 3 and Eq. (23), we obtain that

$$\lim_{j \to \infty} \inf F(z^{k_j}) \geq F(\overline{z}). \tag{24}$$

Choosing $k = k_j - 1$, from Eq. (8), we infer

$$\lim_{j\to\infty} \sup F_i(A_i^{k_j}) \le \lim_{j\to\infty} \sup(F_i(\overline{A_i}) + \frac{1}{2\sigma_i^k}\|\overline{A_i} - Y_i^k\|_F^2 + \langle \nabla_{A_i} H(\{A_n^{k+1}\}_{n=1}^{i-1}, Y_i^k, \{A_n^k\}_{n=i+1}^N), \overline{A_i} - Y_i^k\rangle), \quad (25)$$

where $\lim_{j\to\infty} A_i^{k_j} = \overline{A_i}$. Since $\lim_{j\to\infty} z^{k_j} = \overline{z}$ and $\lim_{j\to\infty} \|z^{k_j} - c^{k_j-1}\|_F = 0$ (Theorem 2(ii)), we can get that

$$\begin{aligned}\lim_{j\to\infty} \|\overline{A_i} - Y_i^k\|_F &\le \lim_{j\to\infty} \|\overline{A_i} - A_i^{k_j}\|_F \\ &+ \lim_{j\to\infty} \|A_i^{k_j} - Y_i^{k_j-1}\|_F \\ &= 0, \end{aligned} \quad (26)$$

From Eq. (25) and Eq. (26), we infer

$$\lim_{j\to\infty} \sup F_i(A_i^{k_j}) \le \lim_{j\to\infty} \sup F_i(\overline{A_i}), \quad (27)$$

where $\lim_{j\to\infty} A_i^{k_j} = \overline{A_i}$. Therefore, from Eq. (23) and Eq. (27), we have

$$\lim_{j\to\infty} \sup F(z^{k_j}) \le \lim_{j\to\infty} \sup F(\overline{z}). \quad (28)$$

From Eq. (23), Eq. (28) and Eq. (24), we infer

$$\lim_{j\to\infty} F(z^{k_j}) = F(\overline{z}). \quad (29)$$

Since Theorem 2, we have

$$\lim_{j\to\infty} H(z^{k_j}) = H(\overline{z}). \quad (30)$$

Thus from Eq. (29) and Eq. (30), we infer

$$\begin{aligned}\lim_{j\to\infty} H(z^{k_j}) + \lim_{j\to\infty} F(z^{k_j}) &= \lim_{j\to\infty} (H(z^{k_j}) + F(z^{k_j})) \\ &= \lim_{j\to\infty} J(z^{k_j}) \\ &= J(\overline{z}).\end{aligned}$$

This means J is constant on z'.

(iii) From Eq. (20), we know that

$$f_i^{k+1} \in \partial_{A_i} J(\{A_j^{k+1}\}_{j=1}^i, \{A_j^k\}_{j=i+1}^N), \quad (31)$$

where the definitions of f_i^{k+1} is the same as in Eq. (18). From Theorem 2(ii) and Lemma 1, we infer

$$\begin{aligned}\lim_{k\to\infty} (\|\{f_i^{k+1}\}_{i=1}^N\|_F &\le \lim_{k\to\infty} \rho_b \|z^{k+1} - c^k\|_F \\ &= 0. \end{aligned} \quad (32)$$

Let

$$z_i^k = A_i^k. \quad (33)$$

From Theorem 2(i), we know that

$$\begin{aligned}\lim_{k\to\infty} J(\left\{z_i^{k+1}\right\}_{i=1}^{j-1}, z_j^{k+1}, \left\{z_i^k\right\}_{i=j+1}^N) &= \lim_{k\to\infty} J(z^k) \\ &= J(z^*). \end{aligned} \quad (34)$$

Therefore, from Eq. (31), Eq. (32), Eq. (33) and Eq. (34), we have

$$\begin{aligned}0 &\in \partial_{z_j} \lim_{k\to\infty} J(\left\{z_i^{k+1}\right\}_{i=1}^{j-1}, z_j^{k+1}, \left\{z_i^k\right\}_{i=j+1}^N) \\ &= \partial_{z_j} J(z^*).\end{aligned}$$

This means $0 \in \partial J(z^*)$. □

E. Proof of Theorem 4

Proof. From Theorem 1, we know that $\frac{\alpha}{2}\operatorname{Tr}((A_N)^T L A_N)$ is an infinitely differentiable function, and the norm of its derivatives of any order is also continuous, so $\frac{\alpha}{2}\operatorname{Tr}((A_N)^T L A_N)$ is a real analytic function (real analytic functions are all KŁ functions [3]). Since the Frobenius norm, ℓ_p-norm and $\ell_{2,1}$-norm are also all KŁ functions [2], it follows that Eq. (6) is a KŁ function. Therefore, from Definition 6, there exists a concave function ϕ so that

$$\phi^{'}(J(z^k) - J(\overline{z}))dist(0, \partial J(z^k)) \ge 1. \quad (35)$$

From ϕ is the convex function, we have

$$\begin{aligned}\phi(J(z^{k+1}) - J(\overline{z})) &\le \phi(J(z^k) - J(\overline{z})) \\ &+ \phi^{'}(J(z^k) - J(\overline{z}))(J(z^{k+1}) - J(z^k)).\end{aligned} \quad (36)$$

From Lemma 1 and Theorem 3, we infer

$$dist(0, \partial J(z^k)) \le \rho_b \|z^k - c^{k-1}\|_F. \quad (37)$$

Since the Eq. (35) and Eq. (37), we infer

$$\begin{aligned}\phi^{'}(J(z^k) - J(\overline{z})) &\ge \frac{1}{dist(0, \partial J(z^k))} \\ &\ge \frac{1}{\rho_b \|z^k - c^{k-1}\|_F}.\end{aligned} \quad (38)$$

Let $J(k) = J(z^k) - J(\overline{z})$, from Eq. (36), Eq. (37), Eq. (38) and Theorem 2(i), we have

$$\begin{aligned}\phi(J(k)) - \phi(J(k+1)) &\ge \phi^{'}(J(k))(J(k) - J(k+1)) \\ &\ge \frac{J(k) - J(k+1)}{\rho_b \|z^k - c^{k-1}\|_F} \\ &\ge \frac{\rho \|z^{k+1} - c^k\|_F^2}{\rho_b \|z^k - c^{k-1}\|_F}.\end{aligned}$$

Define C=$\frac{\rho}{\rho_b}$, C is a constant, so we infer

$$\|z^{k+1} - c^k\|_F^2 \le C(\phi(J(k)) - \phi(J(k+1)))\|z^k - c^{k-1}\|_F.$$

Using the fact that $2ab \le a^2 + b^2$

$$2\|z^{k+1} - c^k\|_F \le C(\phi(J(k)) - \phi(J(k+1))) + \|z^k - c^{k-1}\|_F.$$

Sum both sides

$$\begin{aligned}2\sum_{k=l+1}^K \|z^{k+1} - c^k\|_F &\le \sum_{k=l+1}^K \|z^k - c^{k-1}\|_F \\ &+ C(\phi(J(l+1)) - \phi(J(K+1))) \\ &= C(\phi(J(l+1)) - \phi(J(K+1))) \\ &+ \|z^{l+1} - c^l\|_F + \sum_{k=l+1}^K \|z^{k+1} - c^k\|_F.\end{aligned} \quad (39)$$

From Eq. (39), we can get that

$$\begin{aligned}\lim_{K\to\infty} \sum_{k=l+1}^K \|z^{k+1} - c^k\|_F &\le \|z^{l+1} - c^l\|_F + C\phi(J(l+1)) \\ &- \lim_{K\to\infty} C\phi(J(K+1)) \\ &< \infty.\end{aligned}$$

Thus we have

$$\sum_{k=l+1}^\infty \|z^{k+1} - c^k\|_F < \infty. \quad (40)$$

From Algorithm 1, no matter $c^k = z^k + \beta_k(z^k - z^{k-1})$ or $c^k = z^k$, we always have

$$\|z^{k+1} - z^k\|_F - \beta_k\|z^k - z^{k-1}\|_F \ \leq \|z^{k+1} - c^k\|_F. \quad (41)$$

From Eq. (40) and Eq. (41), we know that

$$\sum_{k=l+1}^{\infty} (\|z^{k+1} - z^k\|_F - \beta_k\|z^k - z^{k-1}\|_F) \leq \sum_{k=l+1}^{\infty} \|z^{k+1} - c^k\|_F < \infty.$$

Since

$$\|z^{k+1} - z^k\|_F - \beta_{max}\|z^k - z^{k-1}\|_F \leq \|z^{k+1} - z^k\|_F - \beta_k\|z^k - z^{k-1}\|_F,$$

we also have

$$\begin{aligned}&\sum_{k=l+1}^{\infty} (\|z^{k+1} - z^k\|_F - \beta_{max}\|z^k - z^{k-1}\|_F)\\ &\leq \sum_{k=l+1}^{\infty} (\|z^{k+1} - z^k\|_F - \beta_k\|z^k - z^{k-1}\|_F)\\ &< \infty, \end{aligned} \quad (42)$$

and

$$\begin{aligned}&\sum_{k=l+1}^{\infty} \|z^{k+1} - z^k\|_F - \sum_{k=l+1}^{\infty} \beta_{max}\|z^k - z^{k-1}\|_F\\ &= \sum_{k=l+1}^{\infty} (\|z^{k+1} - z^k\|_F - \beta_{max}\|z^{k+1} - z^k\|_F)\\ &+ \beta_{max}\|z^{l+1} - z^l\|_F\\ &= \sum_{k=l+1}^{\infty} (1 - \beta_{max})\|z^{k+1} - z^k\|_F\\ &+ \beta_{max}\|z^{l+1} - z^l\|_F. \end{aligned} \quad (43)$$

Let $s^k = \|z^{k+1} - z^k\|_F$, from Eq. (42) and Eq. (43), we have

$$\begin{aligned}\sum_{k=l+1}^{\infty} (1 - \beta_{max})s^k - \beta_{max}s^l &= \sum_{k=l}^{\infty}(s^{k+1} - \beta_{max}s^k)\\ &\leq \sum_{k=l+1}^{\infty} \|z^{k+1} - c^k\|_F.\\ &< \infty.\end{aligned}$$

Thus we infer

$$\sum_{k=l+1}^{\infty} (1 - \beta_{max})s^k < \infty. \quad (44)$$

From $(1 - \beta_{max})$ is a positive constant and Eq. (44), we infer

$$\sum_{k=l+1}^{\infty} \|z^{k+1} - z^k\|_F = \sum_{k=l+1}^{\infty} s^k < \infty.$$

Thus we have

$$\sum_{k=0}^{\infty} \|z^{k+1} - z^k\|_F = \sum_{k=0}^{\infty} s^k < \infty.$$

This shows that

$$\begin{aligned}\lim_{K\to\infty} \|z^{K+p} - z^K\|_F &= \lim_{K\to\infty} \|\sum_{k=K+1}^{K+p} (z^{k+1} - z^k)\|_F\\ &\leq \lim_{K\to\infty} \sum_{k=K+1}^{K+p} \|z^{k+1} - z^k\|_F\\ &= \lim_{K\to\infty} \sum_{k=K+1}^{\infty} s^k\\ &= 0. \end{aligned} \quad (45)$$

This means the Theorem 4 is true. □

Remark 2. *If we let F_i in Eq. (6) be defined as*

$$F_i = \lambda_i\|A_i\|_1 + \mathbb{I}_{A_i\geq 0}(A_i), \quad (46)$$

where $\lambda_i \in \mathbb{R}_+$ ($\forall i \in [N]$). Then we can use the proposed Algorithm 1 to solve ℓ_1-SGDNMF (Eq. (5)), and all theorems in Section V also hold for ℓ_1-SGDNMF. This is because the proof process in Section V only requires that F_i be lower semi-continuous KŁ functions (Definition 3 and Definition 6) with a lower bound, F_i in Eq. (46) satisfy these requirements. The convergence of our proposed algorithms for solving ℓ_1-SGDNMF will also be confirmed with later numerical experiments (see Section X).

F. Proof of Theorem 5

Proof. Let $B_i = \prod_{j=1}^{i-1} A_j$, $C_i = \prod_{j=i+1}^{N} A_j$, the computational complexity of B_i reachs $\mathcal{O}(m_1 \sum_{j=1}^{i-2} m_j m_{j+1})$, and the computational complexity of C_i reachs $\mathcal{O}(R \sum_{j=i+1}^{N-1} m_j m_{j+1})$. Therefore, the computational complexity of $(B_i)^T B_i A_i C_i (C_i)^T$ is

$$\begin{aligned}&\mathcal{O}(R \sum_{j=i+1}^{N-1} m_j m_{j+1} + m_1 \sum_{j=1}^{i-2} m_j m_{j+1}\\ &+ m_i^2 m_{i+1} + m_i m_{i+1}^2) \end{aligned} \quad (47)$$

Considering that B_i and C_i have been pre-calculated, the time complexity of $(B_i)^T X (C_i)^T$

$$\mathcal{O}(m_1 m_i R + m_i m_{i+1} R). \quad (48)$$

Therefore, from Eq. (47) and Eq. (48), we know that the computational cost of $\nabla_{A_i} H(A_{(N)})$ is

$$\begin{aligned}&\mathcal{O}(R \sum_{j=i+1}^{N-1} m_j m_{j+1} + m_1 \sum_{j=1}^{i-2} m_j m_{j+1}\\ &+ m_i^2 m_{i+1} + m_i m_{i+1}^2 + m_1 m_i R + m_i m_{i+1} R). \end{aligned} \quad (49)$$

The cost of computing the Lipschitz constant, projection to nonnegative, and objective evaluation is negligible compared to the cost of computing the partial gradient $\nabla_{A_i} H(A_{(N)})$. Therefore, the time complexity of the Algorithm 1 in each iteration is approximately estimated as

$$\begin{aligned}&\mathcal{O}(\sum_{i=1}^{N-2}(R \sum_{j=i+1}^{N-1} m_j m_{j+1} + m_1) + \sum_{i=3}^{N}(\sum_{j=1}^{i-2} m_j m_{j+1})\\ &+ \sum_{i=1}^{N}(m_i^2 m_{i+1} + m_i m_{i+1}^2 + m_1 m_i R + m_i m_{i+1} R)). \end{aligned} \quad (50)$$

□

X. Supplementary Experimental Results

In this section, we continue to complete the remaining experiments outlined in Section VI.

Table IV: Average and standard deviation results by all methods on the remaining datasets. The best performance is highlighted in bold.

Method \ Dataset	COIL20			ORL		
	ACC	ARI	NMI	ACC	ARI	NMI
$\ell_{2,1}$-NMF	0.589 ± 0.025	0.501 ± 0.017	0.718 ± 0.014	0.503 ± 0.022	0.347 ± 0.025	0.716 ± 0.014
$\ell_{2,1}$-GSNMF	0.609 ± 0.020	0.539 ± 0.020	0.744 ± 0.011	0.542 ± 0.015	0.353 ± 0.025	0.734 ± 0.012
GDNMF	0.524 ± 0.019	0.413 ± 0.020	0.635 ± 0.016	0.418 ± 0.021	0.313 ± 0.024	0.633 ± 0.017
ℓ_1-SDNMF	0.519 ± 0.018	0.428 ± 0.014	0.642 ± 0.014	0.468 ± 0.027	0.321 ± 0.029	0.693 ± 0.017
DHGLpSNMF	0.595 ± 0.011	0.468 ± 0.015	0.688 ± 0.008	0.528 ± 0.004	0.341 ± 0.001	0.677 ± 0.007
ℓ_1-SGDNMF	0.583 ± 0.014	0.452 ± 0.016	0.676 ± 0.013	0.549 ± 0.061	0.378 ± 0.104	0.742 ± 0.062
$\ell_{2,1}$-SSDNMF	0.617 ± 0.011	0.546 ± 0.016	0.735 ± 0.016	0.555 ± 0.012	0.404 ± 0.014	0.759 ± 0.007
$\ell_{2,1}$-SSGDNMF	**0.624 ± 0.010**	**0.553 ± 0.005**	**0.746 ± 0.005**	**0.589 ± 0.010**	**0.452 ± 0.012**	**0.786 ± 0.004**

Method \ Dataset	PNW			OrganSMNIST		
	ACC	ARI	NMI	ACC	ARI	NMI
$\ell_{2,1}$-NMF	0.342 ± 0.034	0.012 ± 0.037	0.135 ± 0.028	0.278 ± 0.008	0.113 ± 0.005	0.270 ± 0.006
$\ell_{2,1}$-GSNMF	0.352 ± 0.033	0.013 ± 0.032	0.139 ± 0.039	0.295 ± 0.008	0.118 ± 0.004	0.263 ± 0.006
GDNMF	0.421 ± 0.011	0.104 ± 0.011	0.290 ± 0.017	0.249 ± 0.006	0.080 ± 0.005	0.186 ± 0.009
ℓ_1-SDNMF	0.424 ± 0.012	0.114 ± 0.009	0.321 ± 0.025	0.229 ± 0.031	0.074 ± 0.027	0.202 ± 0.047
DHGLpSNMF	0.448 ± 0.015	0.121 ± 0.012	0.295 ± 0.016	0.293 ± 0.004	0.101 ± 0.005	0.261 ± 0.001
ℓ_1-SGDNMF	0.451 ± 0.008	0.123 ± 0.006	0.281 ± 0.012	0.299 ± 0.009	0.119 ± 0.007	0.272 ± 0.012
$\ell_{2,1}$-SSDNMF	0.428 ± 0.017	0.118 ± 0.007	**0.330 ± 0.019**	0.240 ± 0.003	0.075 ± 0.015	0.227 ± 0.012
$\ell_{2,1}$-SSGDNMF	**0.467 ± 0.012**	**0.141 ± 0.008**	0.308 ± 0.003	**0.333 ± 0.003**	**0.131 ± 0.002**	**0.316 ± 0.006**

A. Supplementary experiment 1: Clustering performance evaluations on the remaining datasets

We continue to evaluate the clustering performance of all methods on the remaining four real-world datasets described in Section VI, and all parameter settings are the same as in the Section VI. Table IV shows the clustering performance results (average and standard deviation) for the methods over 10 independent runs with different initial points on the remaining datasets. From Table IV, we can get the same conclusion as Section VI as follows.

1) ℓ_1-SGDNMF still outperforms other methods except $\ell_{2,1}$-SSGDNMF and $\ell_{2,1}$-SSDNMF on most datasets. This once again shows that combining the sparsity of factor matrices with graph regularization into DNMF can indeed improve the quality of extracted features.
2) $\ell_{2,1}$-SSDNMF still outperforms both $\ell_{2,1}$-NMF and ℓ_1-SDNMF on most datasets, and $\ell_{2,1}$-SSGDNMF still achieves the best clustering performance on all datasets. These results once again fully demonstrate the effectiveness of our proposed method and show that utilizing the $\ell_{2,1}$-norm to promote the group sparsity of the solution in DNMF can indeed enhance feature extraction capability and interpretability.

B. Supplementary experiment 2: Parameter sensitivity analysis

As outlined in Section VI, we continue to analyze the impact of both the graph regularization parameter α and rank m_{i+1} ($i \in [N-1]$) in $\ell_{2,1}$-SSGDNMF on clustering performance. Specifically, we report the clustering performance under the following two parameter settings on the orlraws10P and Pixraw10P datasets.

1) Fix $N = 3$, $\lambda_i = 1$ ($i \in [N-1]$) and $\alpha = 1$, then alternately vary one of the rank m_{i+1} ($\forall i \in [N-1]$), and record the clustering performance under all possible parameter combinations, as shown in Figure 2(a).
2) Fix $N = 3$, $m_2 = 100$, $m_3 = \lceil \frac{R}{4} \rceil$ and $\lambda_i = 1$ ($i \in [N-1]$), then vary the graph regularization parameter α and record the clustering performance, as shown in Figure 2(b).

From Figure 2, we have the following observations.

1) For the rank, the clustering performance of $\ell_{2,1}$-SSGDNMF on the orlraws10P dataset is relatively stable for small values of m_2 or m_3, but shows a clear negative correlation with both m_2 and m_3 as they increase further. Besides, the clustering performance of $\ell_{2,1}$-SSGDNMF on the Pixraw10P dataset always shows a negative correlation with both m_2 and m_3.
2) For the graph regularization parameter, the clustering performance of $\ell_{2,1}$-SSGDNMF on the orlraws10P dataset tends to stabilize when $\alpha \geq 1$, but the clustering performance of $\ell_{2,1}$-SSGDNMF on the Pixraw10P dataset is relatively insensitive to α.

C. Supplementary experiment 3: Convergence analysis

We also evaluate the convergence performance of our proposed APL algorithm (Algorithm 1) for solving both the ℓ_1-SGDNMF and $\ell_{2,1}$-SSGDNMF. Figure 3 plots how the objective function value changes with respect to time for the APL algorithm on several datasets. From Figure 3, we observe that the objective function value monotonically decreases to a stationary point, thereby validating the effectiveness and convergence of the APL algorithm for solving both the ℓ_1-SGDNMF and $\ell_{2,1}$-SSGDNMF.

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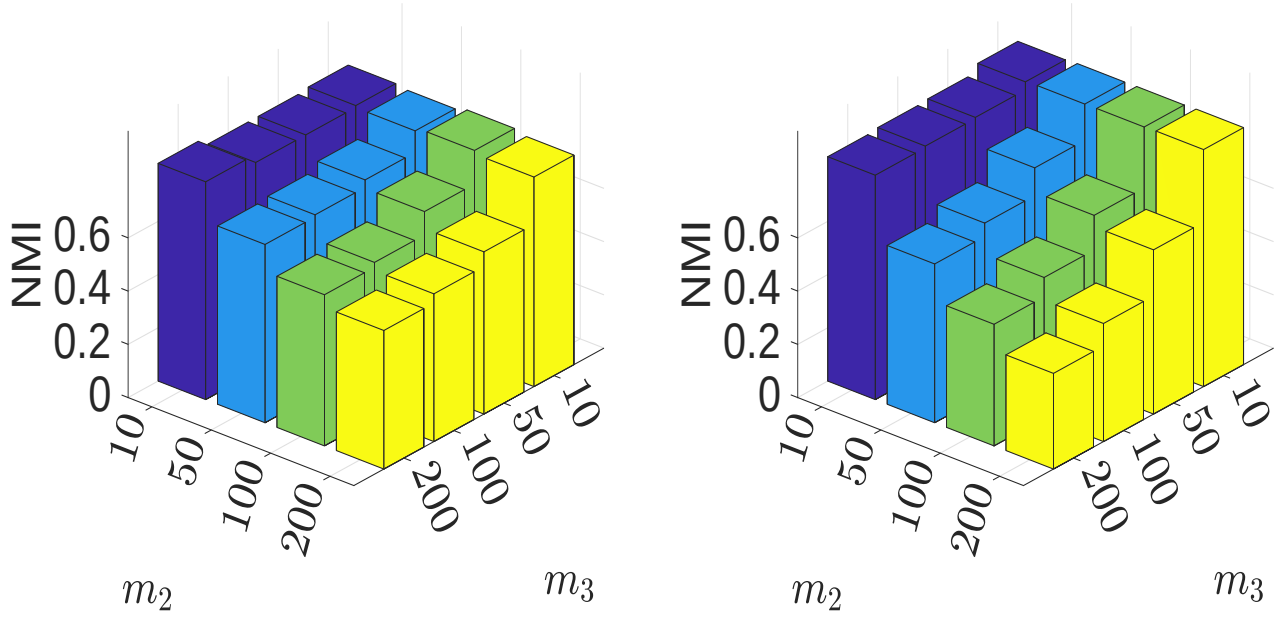
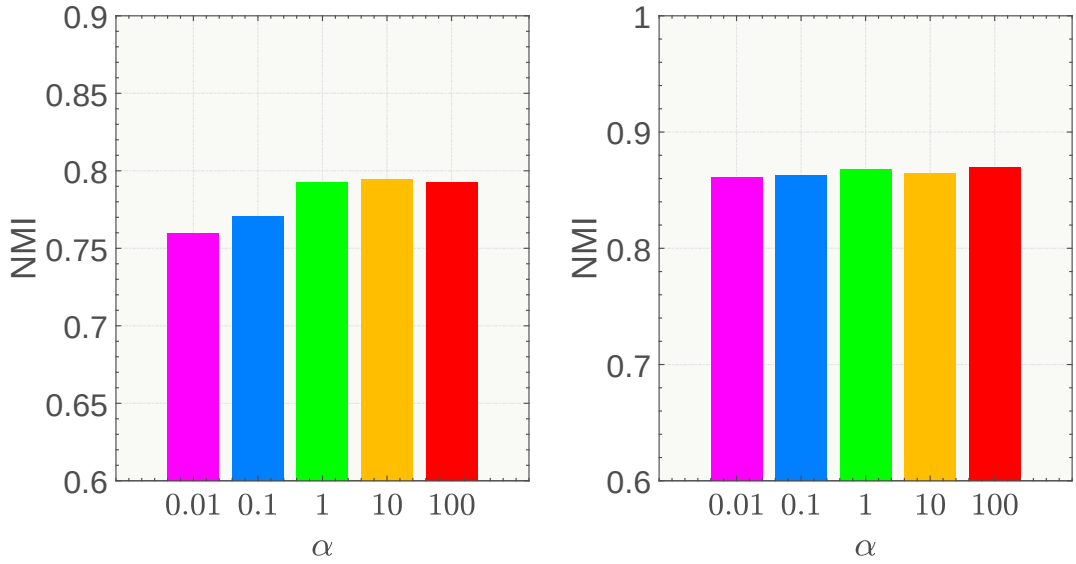


(a) Clustering NMI with respect to the parameters m_2 and m_3 on the orlraws10P (left) and Pixraw10P (right) datasets, respectively.



(b) Clustering NMI with respect to the parameter α on the orlraws10P (left) and Pixraw10P (right) datasets, respectively.

Figure 2: Parameter sensitivity analysis of $\ell_{2,1}$-SSGDNMF on the orlraws10P and Pixraw10P datasets.

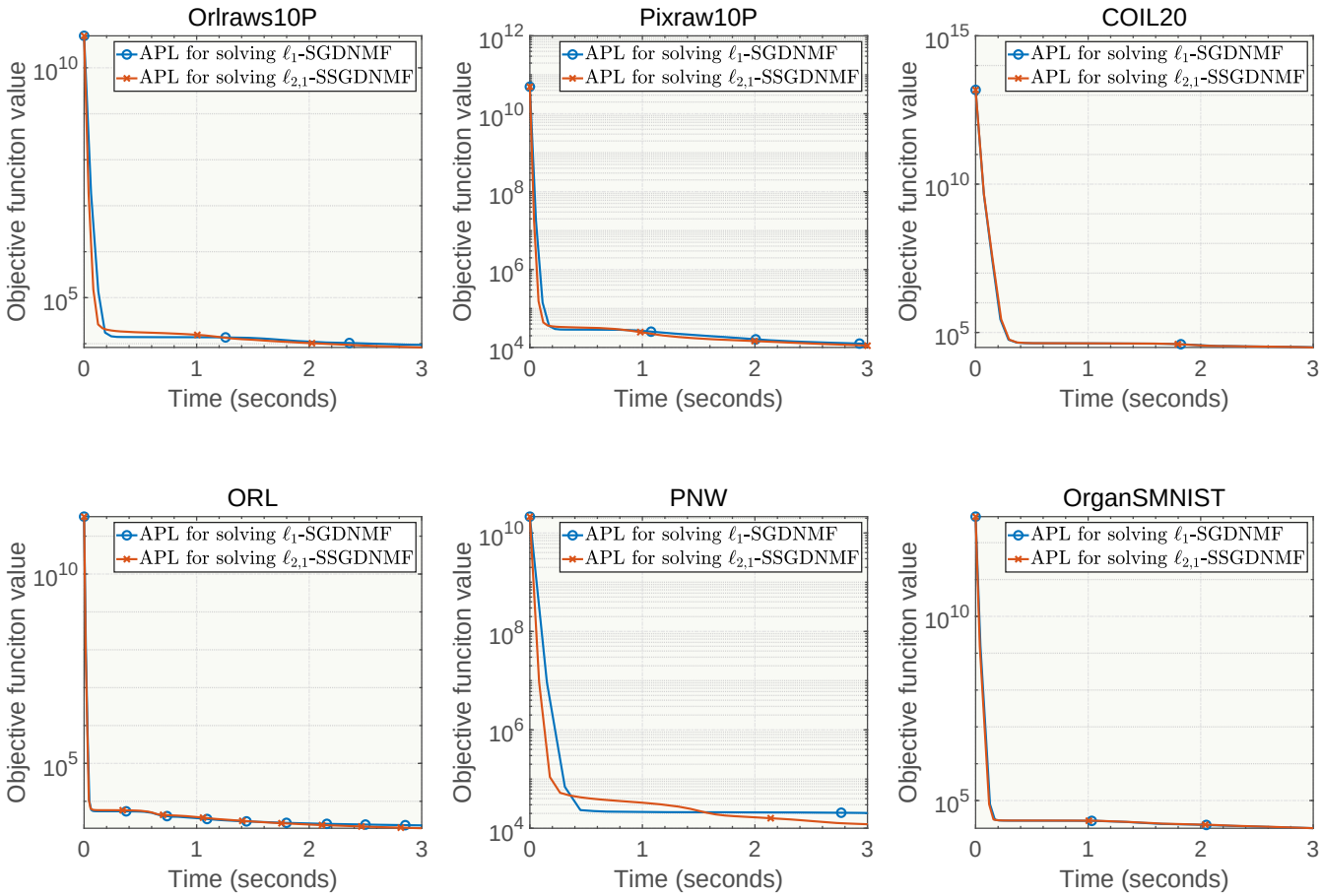


Figure 3: Convergence curves of the APL algorithm for solving both the ℓ_1-SGDNMF and $\ell_{2,1}$-SSGDNMF on six datasets.